

# SOUVIK PODDAR

## Mechanical Engineer | Robotic & Mechatronic Systems | Product Design & Development

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### SUMMARY

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PhD in Mechanical Engineering with 7+ years of experience designing, developing, and validating robotic, electromechanical, and test hardware systems. Core technical strengths include CAD, mechatronic system design, motion control, sensing, mechanism and fixture design, and FEA-based design validation. Hands-on experience includes rapid prototyping, hardware integration, benchtop testing, experimental validation, biomechanics, and motion kinematics. Proven ability to take designs from concept through prototype iteration while executing cross-functionally across mechanical, electrical, controls, and experimental workflows.

### CORE SKILLS

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- Mechanical Design: Component/assembly design, electromechanical subsystem design, mechanism/fixture design
- Hardware Integration: Motor drives, actuators, absolute/incremental encoders, sensors, wiring/harnessing, safety interlocks, modular interfaces
- Controls: Closed-loop control (PID), DAQ, signal processing, multimodal synchronization
- Prototyping & Testing: Test fixtures, benchtop validation, rapid prototyping, machining, 3D printing, laser cutting
- Simulation & Analysis: FEA-driven design validation, nonlinear modeling, Python scripting for FEA
- Biomechanics: Gait analysis, kinematics, motion capture, experimental protocols

### TECHNICAL TOOLS

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- CAD: SolidWorks, Creo, CATIA, Inventor
- FEA: Abaqus/Explicit, ANSYS
- Controls: LabVIEW, Embedded C
- Programming and Data Analysis: Python, MATLAB/Simulink, SPSS
- Motion Capture: Vicon Nexus, OptiTrack Motive, Motion Analysis Cortex

### EXPERIENCE

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#### Research Assistant, University at Buffalo

Buffalo, NY (Aug 2020 - Present)

- Designed and delivered I-PEDLE (Intelligent Prosthesis Emulator for Daily Living Enhancement), a 380 g, 3-DOF robotic platform for evaluating upper-limb prosthesis configurations via adjustable weight and functional modes.
  - Developed CAD models from concept to release: component design, assemblies, electronics mounting, and routing clearances.
  - Engineered a plug-and-play electromechanical architecture integrating NI myRIO controllers, encoders, Hall sensors, motors, load cells, and joystick I/O with serviceable harnessing in a compact structure.
  - Implemented closed-loop motion control and real-time sensing pipelines. Validated via step-response and trajectory-tracking tests (up to 0.7 Hz, under 5% tracking error, no overshoot).
  - Iterated hardware using 3D printing and machining. Built fixtures and test rigs for repeatable verification.
  - Conducted a controlled user study (n=10) across four wrist configurations, quantifying design trade-offs via task performance and upper-limb kinematics.
- Implemented robot-resisted gait training with biofeedback for adults with cerebral palsy (n=5). Improved step length (+9.46%), walking speed (+10.50%), body posture, muscle activation.
  - Led a 4-person cross-functional team and managed robot control and data acquisition.
  - Diagnosed and resolved intermittent runtime issues during live sessions to ensure safe, repeatable operation.
- Teaching Assistant (Aug 2020 - May 2022): Road Vehicle Dynamics, Manufacturing Processes, Intro to ME/AE Practice.
- Mentored master's and undergraduate students on research projects and deliverables.

**Visiting Scholar, University of Michigan****Ann Arbor, MI** (Jan 2025 - Jan 2026)

- Led an I-PEDLE user study (n=15), systematically varying prosthesis weight and quantifying effects on task performance, upper-limb kinematics, and perceived workload (NASA-TLX).
- Designed and 3D-printed study fixtures, built the experimental setup, and developed the end-to-end protocol and data-collection workflow.
- Translated study results into design recommendations by identifying trade-offs between prosthesis weight, functional performance, user workload and fatigue.

**Research Assistant, Arizona State University****Tempe, AZ** (Aug 2018 - May 2020)

- Modeled and optimized inflatable actuator composites using FEA sweeps (33/66/100% volume) and benchtop validation. Achieved ~4.5x faster pressurization.
- Contributed to building a portable pneumatic source and control system (0.131 MPa max pressure, 21.45 SLPM max flow) for wearable-robot operation.
- Designed and characterized a fabric-based soft ankle module to emulate ankle stiffness. Reached ~50% of biological ankle stiffness.
- Developed inflatable insole sensors for gait-phase detection. Fabricated and validated load response up to 700 N.
- Automated Python-driven FEA simulations across soft actuator modes (stiffening, contracting, elongating). Validated via benchtop tests.

**EDUCATION**

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**University at Buffalo****Buffalo, NY**

Doctor of Philosophy, Mechanical Engineering, GPA: 3.80/4.00

May 2026

**Arizona State University****Tempe, AZ**

Master of Science, Mechanical Engineering, GPA: 3.72/4.00

May 2020

**National Institute of Technology****Agartala, India**

Bachelor of Technology, Production Engineering, GPA: 8.34/10

May 2018

**SELECTED PUBLICATIONS**

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- Poddar et al. (2026). A Robotic Emulator for User-Driven Design of Multi-DOF Transradial Prostheses. IEEE Robotics and Automation Letters. DOI: [10.1109/LRA.2026.3664529](https://doi.org/10.1109/LRA.2026.3664529)
- Poddar et al. (2025). Improving Foot-Rocker via Robot-Resisted Gait Training with Self-awareness Biofeedback in Adults with Cerebral Palsy. IEEE Transactions on Neural Systems and Rehabilitation Engineering. DOI: [10.1109/TNSRE.2025.3636432](https://doi.org/10.1109/TNSRE.2025.3636432)
- Poddar et al. (2022). A lightweight transradial prosthetic emulator for optimizing prosthetic wrist design. IEEE International Conference on Biomedical Robotics and Biomechatronics. DOI: [10.1109/BioRob52689.2022.9925286](https://doi.org/10.1109/BioRob52689.2022.9925286)
- All publications: <https://scholar.google.com/citations?user=HyodwH8AAAAJ&hl=en&oi=ao>

**PATENTS, AWARDS AND FELLOWSHIPS**

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- U.S. Patent US 12,458,556 B2 (Issued 2025): Low-volume inflatable actuator composites for wearable applications and portable pneumatic source.
- Research supported by the National Science Foundation Disability and Rehabilitation Engineering (DARE) program award (Aug 2022 - Present).
- Master's Opportunity for Research in Engineering (MORE) award (Aug 2019 - May 2020).